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|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---|----------|---|---------|---|---------|
| 7  | A uniform cube of volume $0.729 \text{ m}^3$ is floating in water. The density of water is $1000 \text{ kg m}^{-3}$ .<br>A load of $400 \text{ N}$ is then placed onto the cube. The cube remains afloat.<br>What is the change in the depth of the cube submerged in the water after the load is added?                                                        |          |   |          |   |         |   |         |
|    | A                                                                                                                                                                                                                                                                                                                                                               | 0.0503 m | B | 0.0559 m | C | 0.494 m | D | 0.900 m |
| L2 | <b>Answer: A</b><br><br>Let the change in depth be $d$ ,<br><br>Cross-sectional area of the cube $= \left(\sqrt[3]{0.729}\right)^2 = 0.81 \text{ m}^2$<br><br>By the principle of flotation,<br>Weight of <b>added load</b> = <b>Additional</b> weight of the water displaced due to the load's weight<br>$400 = (0.81d)(1000)(9.81)$<br>$d = 0.0503 \text{ m}$ |          |   |          |   |         |   |         |

[Turn over

8 A right-angle rule hangs at rest from a peg P as shown below. The rule is uniform in density